

Prop Let $T : V \rightarrow W, r := \text{rk} T$. From SVD we can get ONBs $\underline{v}, \underline{w}$ and singular values σ_i . Define $S : W \rightarrow V$,

$$Sw_j = \begin{cases} \sigma_j^{-1} v_j, & 1 \leq j \leq r \\ 0, & j > r \end{cases}$$

Then S is an MP inverse of T .

In terms of matrices, $\exists P \in M_{m \times m}(\mathbb{R}), Q \in M_{n \times n}(\mathbb{R})$, both orthogonal, $Q^{-1}AP = \text{diag}(\sigma_i)$. Define $S \leftrightarrow$ matrix B such that $P^{-1}BQ = \text{diag}(\sigma_i^{-1})$.

Proof Use the explicit definition construction of MP inverse $Sw = v''$.

$$\begin{aligned} \forall w \in W, w &= \sum_{j=1}^n a_j w_j, \text{im } T = \langle w_1, \dots, w_r \rangle \\ \implies w' &= \sum_{j=1}^r a_j w_j, \text{ Set } v := \sum_{j=1}^r \sigma_j^{-1} a_j v_j, Tv = w' \\ v &\in \langle v_{r+1}, \dots, v_m \rangle^\perp = (\ker T)^\perp \\ \implies v &= 0 + v = v' + v'' \end{aligned}$$

Therefore the definition of S agrees with the construction of MP inverse of T . ■

§ Min-Max principle

Let V be an inner product space on $\mathbb{R}$. Let B be a symmetric bilinear form $V \times V \rightarrow \mathbb{R}$. Then $\exists! S \in \text{End}(V), S^* = S$ such that $B(v_1, v_2) = (v_1 | Sv_2)$. In fact, we can identify V with $\mathbb{R}^n + \text{std IP}$.

Then we observe that

$$\exists! A \in M_{n \times n}(\mathbb{R}), B(v_1, v_2) = v_1^T A v_2 = v_1 \cdot A v_2.$$

Therefore S is the A in the equation above, and S is self-adjoint since B is symmetric.

$\exists$ ONB $\underline{v}$ of $V, \lambda_1 \geq \dots \geq \lambda_n$ such that $Sv_i = \lambda_i v_i$.

A simple observation:

$$\lambda_1 = \max_{\|v\|=1} B(v, v), \lambda_n = \min_{\|v\|=1} B(v, v).$$

Proof Let $v = a_1 v_1 + \dots + a_n v_n, \|v\| = 1 \iff \sum_{i=1}^n a_i^2 = 1$.

$$\lambda_n \leq \left(B(v, v) = (v | Sv) = \sum_{i=1}^n a_i^2 \lambda_i \right) \leq \lambda_1. \blacksquare$$

Remark

- $\{v \in V | \|v\| = 1\}$ is compact (if $V \simeq \mathbb{R}^n$) and $B(v, v)$ is continuous $\implies$ extrema exists.
- Given $S \in \text{End}(V), S = S^*$, can show $\max_{\|v\|=1} (v | Sv)$ attained at v : eigenvector of S (Set $V = \mathbb{R}^n$ and use Lagrange multipliers). Can get another real proof of diagonalizable theorem.
- $\max_{\|v\|=1} B(v, v) = \max_{v \neq 0} \frac{B(v, v)}{(v|v)}$, since $B(tv, tv) = t^2 B(v, v), (tv|tv) = t^2 (v|v)$.

Theorem(Courant-Fischer) Given $S : V \rightarrow W, S = S^*, \forall 1 \leq k \leq n := \dim V :$

$$\lambda_k = \min_{\substack{U \subset V, \\ \dim U = n-k+1}} \left\{ \max_{\|v\|=1, v \in U} B(v, v) \right\},$$

$$\lambda_k = \max_{\substack{U \subset V, \\ \dim U = k}} \left\{ \min_{\|v\|=1, v \in U} B(v, v) \right\}.$$

Proof Replace S by $-S$: $\min \leftrightarrow \max$. Suffices to prove the first equation.

Assume $U \in V, \dim U = n - k + 1$.

$$\begin{aligned} \dim U \cap \langle v_1, \dots, v_k \rangle &= \dim U + k - \dim U \cup \langle v_1, \dots, v_k \rangle \\ &\geq \dim U + k - n \\ &= 1. \end{aligned}$$

So we can take $v = \sum_{i=1}^k a_i v_i \in U, \|v\| = 1$.

$$B(v, v) = (v|Sv) = \sum_{i=1}^k \lambda_i a_i^2 \geq \lambda_k \sum_{i=1}^k a_i^2 = \lambda_k.$$

$$\max_{\|v\|=1, v \in U} B(v, v) \geq \lambda_k \implies \inf_{\dim U = n-k+1} \left\{ \max_{\|v\|=1, v \in U} B(v, v) \right\} \geq \lambda_k.$$

To show that minimum can be attained, take $U = \langle v_k, \dots, v_n \rangle. \forall v = \sum_{i=k}^n a_i v_i, \|v\| = 1$,

$$B(v, v) = \sum_{i=k}^n \sigma_i a_i^2 \leq \lambda_k \sum_{i=k}^n a_i^2 = \lambda_k.$$

Therefore $\max_{\|v\|=1, v \in U} B(v, v) = \lambda_k$ for this U . ■

This is also related to SVD.

Proposition Let V, W be IPSs on $\mathbb{R}$ with $\dim < \infty$. Let T be a linear map $V \rightarrow W$ with singular values $\sigma_1 \geq \dots \geq \sigma_i \geq \dots$ and set $\sigma_i = 0$ when $i > \max\{\dim V, \dim W\}$. Then

$$\sigma_1 = \max_{v \neq 0} \frac{\|Tv\|}{\|v\|}, \quad \sigma_{\dim V} = \min_{v \neq 0} \frac{\|Tv\|}{\|v\|}.$$

Proof Take $S = T^*T \in \text{End}(V), S = S^*$ and let B be the bilinear form corresponding to S .

$$B(v_1, v_2) = (v_1|Sv_2) = (Tv_1|Tv_2), B(v, v) = \|Tv\|^2, \frac{B(v, v)}{(v|v)} = \left(\frac{\|Tv\|}{\|v\|} \right)^2.$$

Recall that $\sigma_1^2 \geq \dots \geq \sigma_{\dim V}^2$ are the eigenvalues of S . From the simple observation above we proved this proposition. ■

§ Positive matrices

Let $A \in M_{m \times n}(\mathbb{R}), A \geq B$ means $\forall i, j, a_{ij} \geq b_{ij}$. $A > 0, A \geq 0$ is defined similarly, not to be mistaken by positive- or semi-positive definite.

Lemma $\forall A \in M_{m \times n}(\mathbb{R}), x \in \mathbb{R}^n = M_{n \times 1}(\mathbb{R})$.

- $A > 0, x \geq 0, x \neq 0 \implies Ax > 0$,
- $A > 0, x \geq 0 \implies Ax \geq 0$.

Proof The conclusion is obvious. ■

To study the eigenvalues and eigenvectors of positive matrices $A \in M_{m \times n}(\mathbb{R})$, $A > 0$, we give the following definition:

Definition For $A \in M_{m \times n}(\mathbb{R})$, its **spectral radius** $\rho(A) = \max_{\lambda \in \mathbb{C}, \text{ eigenvalue}} |\lambda|$.

Theorem (Perron) Let $A \in M_{n \times n}(\mathbb{R})$, $A > 0$.

1. $\rho(A) > 0$, $\exists v \in \mathbb{R}^n$, $v > 0$, $Av = \rho(A)v$.
2. If μ : eigenvalue of A satisfies $\mu \neq \rho(A)$, then $|\mu| < \rho(A)$.
3. The $\rho(A)$ -eigenspace of A has $\dim = 1$.
4. $\rho(A)$ is a single root of Char_A .

After adding more conditions, can get a more general version for $A \geq 0$ (Perron-Frobenius Theorem).

Lemma $A \in M_{n \times n}(\mathbb{R})$, $A > 0 \implies \exists v \in \mathbb{R}^n$, $v > 0$, $Av = \rho(A)v$.

Proof Let $S := \{x \in \mathbb{R}^n \mid \|x\| = 1, x \geq 0\}$. Consider the continuous map

$$L : S \rightarrow \mathbb{R}_{>0}$$

$$x \mapsto \min_{1 \leq i \leq n, x_i \neq 0} \frac{(Ax)_i}{x_i}$$

Since S is compact, $\exists v \in S$, s.t. L attains its maximum $\rho > 0$ at v . Here we claim that

$$Av = \rho v \wedge v > 0.$$

From $L(v) = \rho$ we know $Av \geq \rho v$. If $Av \neq \rho v$, $Av - \rho v \geq 0$, then $A(Av - \rho v) > 0$.

Therefore for some small ε ,

$$A(Av - \rho v) \geq \varepsilon Av \implies A^2 v \geq (\varepsilon + \rho)Av \implies Aw \geq (\varepsilon + \rho)w$$

where $w = tAv$, $t \in \mathbb{R}$, $w \in S$. Hence we reach a contradiction $\implies Av = \rho v$.

$v = \rho^{-1} \underbrace{Av}_{>0 \neq 0} \implies v > 0$. Also from definition of $\rho(A)$ we have $\rho \leq \rho(A)$.

To show that $\rho = \rho(A)$, let $\mu \in \mathbb{C}$, $w \in \mathbb{C}^n \setminus \{0\}$ s.t. $Aw = \mu w$.

For this equation we can show $\forall i, |\mu|w_i = |(\mu w)_i|$

$$= \left| \sum_{j=1}^n a_{ij}w_j \right| \leq \sum_j a_{ij}|w_j|.$$

Set $w' = (|w_1|, \dots, |w_n|) \in \mathbb{C}^n$, and scale it to ensure $\|w'\| = 1$.

$w' \geq 0$, $Aw' \geq \mu w'$, $w' \in S \implies \rho(A) \geq \rho \geq L(w') \geq |\mu|$ for all eigenvalues $\mu \in \mathbb{C}$. Hence $\rho = \rho(A)$. ■

Proof of Theorem (1) done. (2) continue the previous proof.

We already showed $\rho(A) = \max_{v \in S} L(v) \geq L(w') \geq |\mu|$, and we need to prove $|\mu| = \rho(A) \implies \mu = \rho(A)$. We reason that $|\mu| = \rho(A) \implies L(w') = \rho(A)$ from the inequality above, therefore

$$Aw' = \rho(A)w' \implies \forall i, \sum_j a_{ij}|w_j| = \rho(A)w_i = |\mu w_i| = \left| \sum_j a_{ij}w_j \right|$$

Hence all w_i are on the same ray in $\mathbb{C}$. Take some nonzero number c on this ray and $\forall i, c^{-1}w_i \in \mathbb{R}$, $c^{-1}w_i \geq 0$. This shows that μ is a real number. ■